



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 2

Experiment Title: Design, Implementation, and Simulation of a 1-Bit Full Adder Using Xilinx ISE

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1 Experiment Name

Design, Synthesis, and Functional Simulation of a 1-Bit Full Adder Circuit using Verilog HDL in Xilinx ISE Design Suite.

2 Objective

- To design a 1-bit Full Adder module using continuous dataflow modeling in Verilog HDL.
- To synthesize the design using Xilinx Synthesis Technology (XST) and extract the top-level block and gate-level RTL schematics.
- To verify the functional correctness of the full adder logic across all eight possible input permutations via testbench simulation using the ISim simulator.

3 Apparatus / Hardware & Software Requirements

- **Host PC:** Windows 10 (x64 architecture)
- **Virtualization Platform:** Oracle VM VirtualBox
- **Operating System Environment:** Oracle Linux 64-bit VM
- **EDA Software Suite:** Xilinx ISE Design Suite 14.7
- **Target Architecture:** Xilinx Artix-7 (xc7a100t-3csg324)
- **Simulation Tool:** ISim (VHDL/Verilog)

4 Circuit Diagram / Schematic

The Boolean logic equations implemented for the 1-bit Full Adder are expressed as:

$$Sum = A \oplus B \oplus Cin$$

$$Cout = (A \cdot B) + (Cin \cdot (A \oplus B))$$

4.1 Top-Level Block RTL Schematic

The top-level symbol defines three single-bit inputs (A , B , Cin) and two single-bit outputs (Sum , $Cout$).

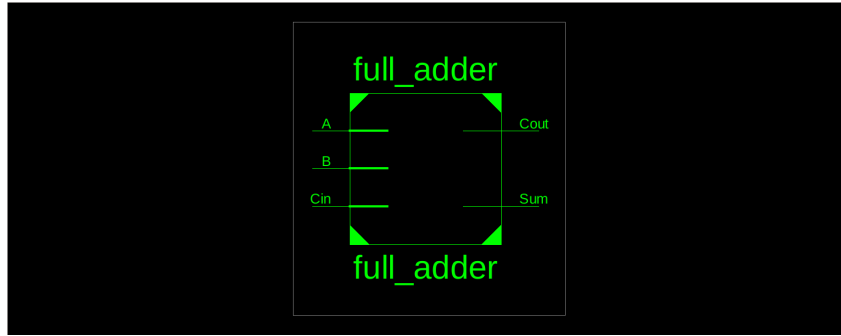


Figure 1: Top-Level RTL Block Schematic of the 1-bit Full Adder.

4.2 Gate-Level RTL Schematic

Synthesizing the module expands the internal register-transfer logic into two XOR gates (`Mxor_A_B_XOR_1_o1` and `Mxor_Sum1`), two AND gates (`and2`), and an OR gate (`or3`) mapped to drive the *Sum* and *Cout* outputs.

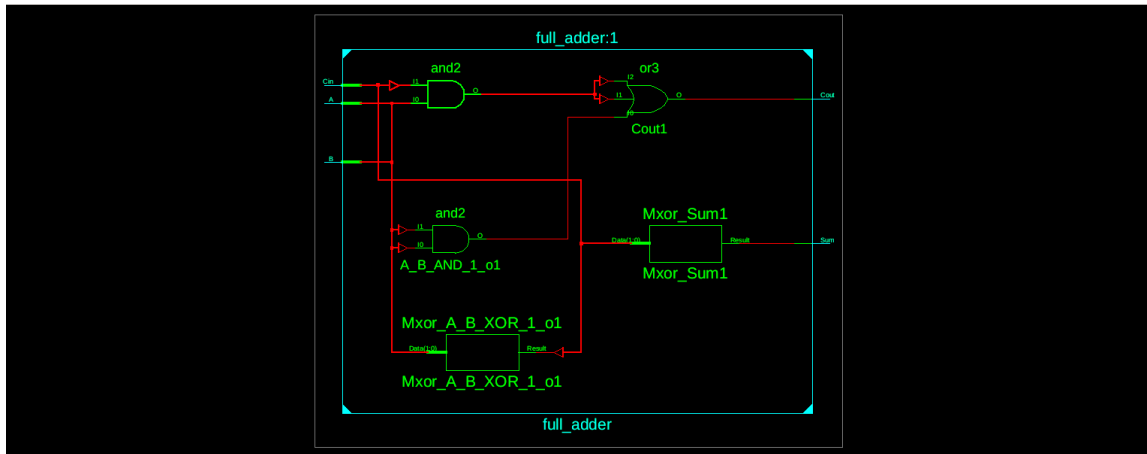


Figure 2: Internal Synthesized Gate-Level RTL Schematic.

5 Output Analysis / Observations

The behavioral simulation was executed using a Verilog Test Fixture (`full_adder_tb.v`) configured with 10 ns delays for each consecutive binary input sequence spanning from 000 to 111, for a total observation period of 80 ns.

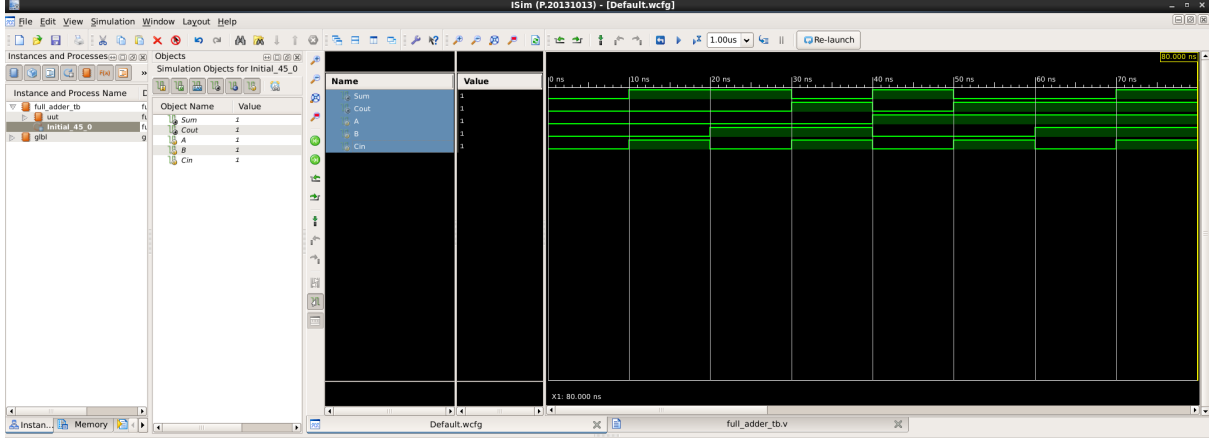


Figure 3: ISim Behavioral Simulation Waveform across 0 ns to 80 ns.

5.1 Waveform Verification Truth Table

Table 1: Verification Table Comparing Input Stimuli to Output Logic States

Time Interval	Input A	Input B	Input C_{in}	Expected Output (C_{out}, Sum)	Simulated C_{out}	Simulated Sum
0 ns – 10 ns	0	0	0	00	0	0
10 ns – 20 ns	1	0	0	01	0	1
20 ns – 30 ns	1	1	0	10	1	0
30 ns – 40 ns	1	1	1	11	1	1
40 ns – 50 ns	0	1	1	10	1	0
50 ns – 60 ns	0	0	1	01	0	1
60 ns – 70 ns	0	1	0	01	0	1
70 ns – 80 ns	1	1	0	10	1	0

6 Result

- The 1-bit Full Adder was synthesized successfully without syntax errors or warnings using XST.
- Functional behavioral simulation in ISim strictly matched the theoretical truth table for all eight combinations without timing discrepancies.

- The primary objective of the experiment was achieved.

7 Github Repository Commits

- **Repository Link:** <https://github.com/IstiakNabil/Vlsi-Lab-Report>
- **Committed Files:**
 - `full_adder.v` (Design Module)
 - `full_adder_tb.v` (Testbench Module)

8 Conclusion

In this experiment, a digital design workflow was executed using Xilinx ISE Design Suite 14.7 within an Oracle Linux VirtualBox environment. The full adder was modeled using dataflow Verilog constructs, synthesized into an optimized gate netlist, and verified functionally with the ISim waveform viewer. The simulated waveform and synthesized schematic conformed to standard digital logic and binary addition principles.